MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL,
COUNTERS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of Military Specification MIL-M-38510/315C, dated 17 January 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Title: Delete and substitute as printed above.

1.2.3, Case outline E: Delete "14-lead" and substitute "16-lead".

PAGE 2

1.3, Thermal resistance, delete and substitute:

"*Thermal resistance, junction-to-case* (θ_{JC}):
Cases A, B, C, D, E, and F - - - - - - - - - - (see MIL-M-38510, appendix C)
Case 2 - 60°C/W 3/

1.3, Junction temperature, delete and substitute:

"Junction temperature (TJ) 4/ - - - - - - - - - +175°C"

Following footnote 2/, add new footnotes as follows:

"3/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

4/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

* 1.4, Supply voltage: Add "(V_{CC})".

* 3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

PAGE 4

* TABLE I, I_{IL3}, delete "I_{IL3}" and substitute as below:

	Device type	Limits Min	Max	Units
I_{IL3}	01, 10	-0.4	-3.2	mA
	02	-0.4	-1.6	mA

PAGE 6

TABLE I, F_{MAX}, t_{PHL1}, t_{PLH1}, t_{PHL2}, t_{PLH2}, t_{PHL4}, t_{PLH4}, conditions column: After "C_L = 50 pF", add "±10%".

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PAGES 7 AND 8

TABLE I, delete and substitute new table I as printed on pages 4 and 5 of this amendment.

PAGE 9

3.6, second sentence, delete in its entirety.

4.2a, (1) After test condition D add: "or E".

PAGE 38

FIGURE 2, delete and substitute new figure 2 as printed on page 6 of this amendment.

PAGE 81

Footnote 7/, I_{IL1} parameter, circuit C: Delete "-120/-360" and substitute "-.12/-.36".

PAGES 95 THROUGH 101

TABLE III, pin designation 2 column: Delete "CP" and substitute "CK".

PAGES 102 AND 103

TABLE III, delete and substitute new table III as printed on pages 7 and 8 of this amendment.

PAGE 104

TABLE III, footnote 3/, I_{IL15}, Circuit C, delete "-290/-720" and substitute "-140/-720".

PAGE 105

* TABLE III, VoL, Test 5 and 6 pin 16: Add "4.5 V".

PAGE 113

* TABLE III, note 3, Circuit B: Delete "(08 only)".

PAGES 113 AND 122

TABLE III, footnote 4/, delete "B" and substitute "B = -15/-110".

PAGE 124

TABLE III, subgroup 9, t_{PHL2} Test 70 Max Limits: Delete "40 ns" and substitute "50 ns".

PAGE 126

4.4.3b, delete in its entirety.

4.4.3, c (1) after test condition D: Add "or E".

PAGE 127

6.1, delete in its entirety.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0806)

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
Propagation delay time, low to high, clock to Q	t_{PLH5}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 2 kΩ	03, 04, 11, 12	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		03, 04, 11, 12	3	45	ns
Propagation delay time, low to high, clock to Q	t_{PLH5}		05, 06	3	26	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		05	3	26	ns
			06	3	36	ns
Propagation delay time, low to high, clock (data) to Q	t_{PLH6}		03, 04, 11, 12	3	42	ns
Propagation delay time, high to low, clock (data) to Q	t_{PHL6}		03, 04, 11, 12	3	48	ns
Propagation delay time, low to high, EnT to carry	t_{PLH7}		03, 04, 11, 12	3	28	ns
Propagation delay time, high to low, EnT to carry	t_{PHL7}		03, 04, 11, 12	3	28	ns
Propagation delay time, low to high, ET to RC	t_{PLH7}		05	3	18	ns
			06	3	28	ns
Propagation delay time, high to low, ET to RC	t_{PHL7}		05	3	28	ns
			06	3	32	ns
Propagation delay time, high to low, clear to Q	t_{PHL8}		03, 04, 11, 12	3	46	ns
Propagation delay time, low to high, load to Q	t_{PLH8}		07, 08	3	63	ns
Propagation delay time, high to low, load to Q	t_{PHL10}		07, 08	3	63	ns
Propagation delay time, low to high, counts up and down to Q, U/$\overline{D}$ to RC	t_{PLH9}		07, 08	3	60	ns
			05	3	26	ns
			06	3	32	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
Propagation delay time, high to low, counts up and down to Q, U/$\overline{D}$ to RC	t_{PHL11}	V_{CC} = 5.0 V, C_L = 50 pF, ±10%, R_L = 2 kΩ	07, 08	3	73	ns
			05	3	33	ns
			06	3	37	ns
Propagation delay time, high to low, clear to Q	t_{PHL12}		07, 08	3	56	ns
Propagation delay time, low to high, load to Q	t_{PLH10}		09, 13	3	53	ns
Propagation delay time, high to low, load to Q	t_{PHL13}		09, 13	3	77	ns
Propagation delay time, low to high, clock to Q	t_{PLH11}		09, 13	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL14}		09, 13	3	57	ns
Propagation delay time, low to high, clock to Max/Min	t_{PLH12}		09, 13	3	66	ns
Propagation delay time, low to high, clock to ripple carry	t_{PLH12}		05	3	35	ns
			06	3	38	ns
Propagation delay time, high to low, clock to Max/Min	t_{PHL15}		09, 13	3	80	ns
Propagation delay time, high to low, clock to ripple carry	t_{PHL15}		05	3	37	ns
			06	3	40	ns

1/ Use I_{OL} + $I_{IL3(MAX)}$ for V_{OL} test on Q_A.

2/ Not more than one output should be shorted at a time.

3/ I_{CCH} is measured: (a) With the load input high; and (b) Then again with the load input low with all other inputs high and all outputs open.

4/ I_{CCL} is measured: (a) With the clock input high; and (b) Then again with the clock input low with all other inputs low and all outputs open.

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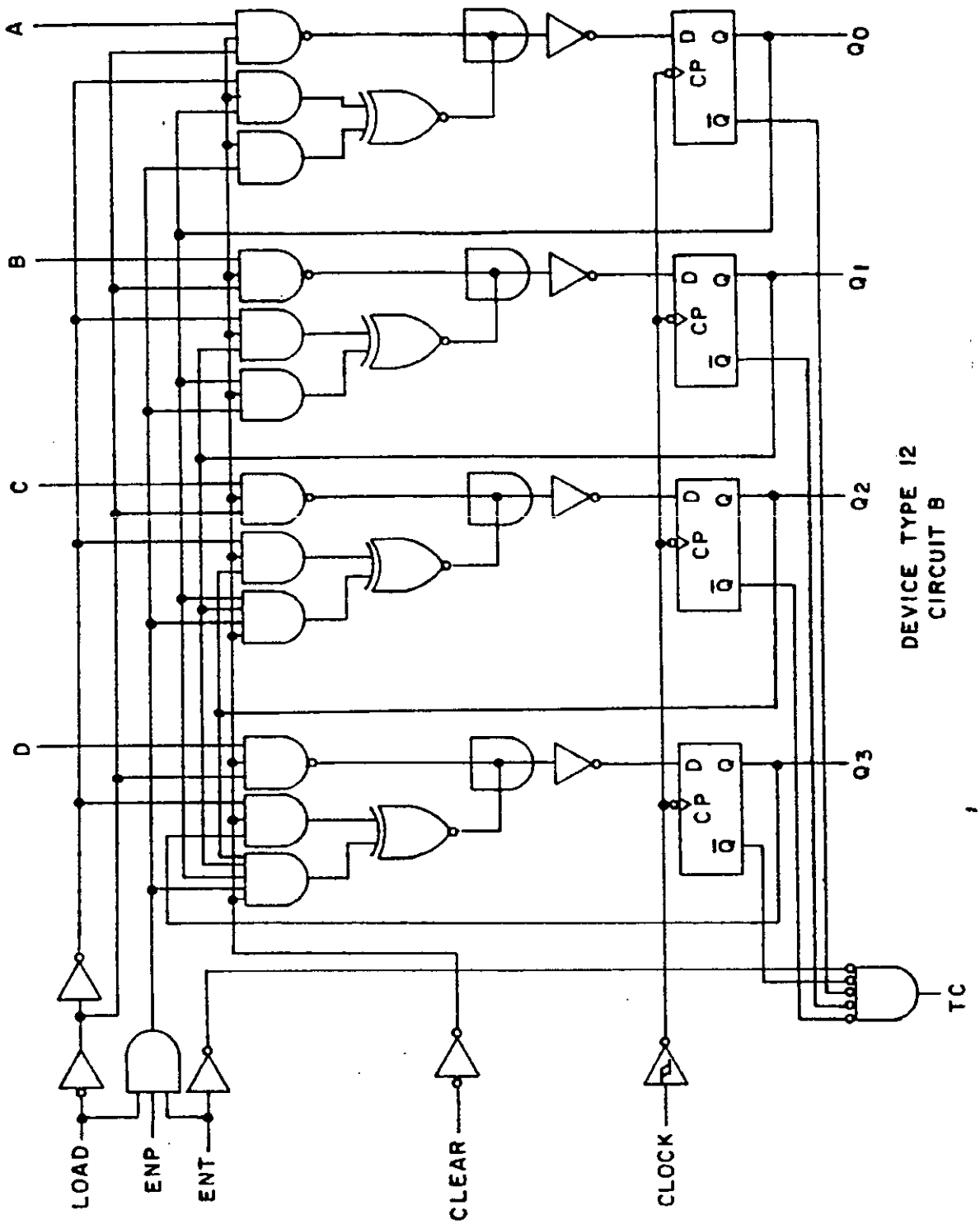


FIGURE 2. Logic diagram - Continued.

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TABLE III. Group A inspection for device type 06.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 / U/$\overline{D}$	2 / CK	3 / A	4 / B	5 / C	6 / D	7 / EP	8 / GND	9 / L	10 / ET	11 / Q_D	12 / Q_C	13 / Q_B	14 / Q_A	15 / Ripple carry	16 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PLH5}	See fig. 7	129	5.0 V	IN 7/	GND	GND	GND	GND	GND	GND	IN	GND				OUT		5.0 V	CP to Q_A	3	22	ns
			130	"	"	5.0 V	"	"	"	"	"	"	"			OUT			"	CP to Q_B	"	"	"
			131	"	"	"	5.0 V	"	"	"	"	"	"		OUT				"	CP to Q_C	"	"	"
			132	"	"	"	"	5.0 V	"	"	"	"	"	OUT					"	CP to Q_D	"	"	"
	t_{PHL5}	"	133	"	" 7/	5.0 V	GND	GND	GND	"	GND	IN	GND				OUT		"	CP to Q_A	"	32	"
			134	"	"	"	5.0 V	"	"	"	"	"	"			OUT			"	CP to Q_B	"	"	"
			135	"	"	"	"	5.0 V	"	"	"	"	"		OUT				"	CP to Q_C	"	"	"
			136	"	"	"	"	"	5.0 V	"	"	"	"	OUT					"	CP to Q_D	"	"	"
	t_{PHL15}	"	137	"	"	GND	"	"	"	"	"	"	"					OUT	"	CP to RC	"	35	"
	t_{PLH12}	"	138	"	"	GND	"	"	"	"	"	5.0 V	"					OUT	"	CP to RC	"	33	"
	t_{PLH5}	"	139	GND	"	5.0 V	GND	GND	GND	"	"	GND	"				OUT		"	CP to Q_A	"	22	"
	t_{PHL5}	"	140	"	"	GND	GND	"	"	"	"	"	"				OUT		"	CP to Q_A	"	32	"
	t_{PLH5}	"	141	"	"	"	5.0 V	"	"	"	"	"	"			OUT			"	CP to Q_B	"	22	"
	t_{PHL5}	"	142	"	"	"	GND	"	"	"	"	"	"			OUT			"	CP to Q_B	"	32	"
	t_{PLH5}	"	143	"	"	"	"	5.0 V	"	"	"	"	"		OUT				"	CP to Q_C	"	22	"
	t_{PHL5}	"	144	"	"	"	"	GND	"	"	"	"	"		OUT				"	CP to Q_C	"	32	"
	t_{PLH5}	"	145	"	"	"	"	"	5.0 V	"	"	"	"	OUT					"	CP to Q_D	"	22	"
	t_{PHL5}	"	146	"	"	"	"	"	GND	"	"	"	"	OUT					"	CP to Q_D	"	32	"
	t_{PHL7}	"	147	5.0 V	" 7/	5.0 V	5.0 V	5.0 V	5.0 V	"	"	5.0 V	IN					OUT	"	ET to RC	"	28	"
	t_{PLH7}	"	148	5.0 V	"	"	"	"	"	"	"	5.0 V	IN					OUT	"	ET to RC	"	24	"
	t_{PHL11}	"	149	IN	IN 7/	"	"	"	"	"	"	5.0 V	GND					OUT	"	U/$\overline{D}$ to RC	"	32	"
	t_{PLH9}	"	150	IN	IN	"	"	"	"	"	"	5.0 V	GND					"	"	U/$\overline{D}$ to RC	"	28	"
	t_{PHL7}	"	151	GND	IN 7/	GND	GND	GND	GND	"	"	5.0 V	IN					"	"	ET to RC	"	28	"
	t_{PLH7}	"	152	GND	IN	"	"	"	"	"	"	5.0 V	IN					"	"	ET to RC	"	24	"
	t_{PHL11}	"	153	IN	IN 7/	"	"	"	"	"	"	5.0 V	GND					"	"	U/$\overline{D}$ to RC	"	32	"
	t_{PLH9}	"	154	IN	IN	"	"	"	"	"	"	5.0 V	"					"	"	U/$\overline{D}$ to RC	"	28	"
	F_{MAX}	"	155	5.0 V	IN	"	1/	"	"	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz
	F_{MAX}	"	156	GND	IN	5.0 V	5.0 V	5.0 V	5.0 V	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/D	CX	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
10 T_C = +125°C	t_{PLH5}	See fig. 7	157																	CP to Q_A	3	26	ns
		"	158																	CP to Q_B	"	"	"
		"	159																	CP to Q_C	"	"	"
		"	160																	CP to Q_D	"	"	"
	t_{PHL5}	"	161																	CP to Q_A	"	36	"
		"	162																	CP to Q_B	"	"	"
		"	163																	CP to Q_C	"	"	"
		"	164																	CP to Q_D	"	"	"
	t_{PHL15}	"	165																	CP to RC	"	40	"
	t_{PLH12}	"	166																	CP to RC	"	38	"
	t_{PLH5}	"	167																	CP to Q_A	"	26	"
	t_{PHL5}	"	168																	CP to Q_A	"	35	"
	t_{PLH5}	"	169																	CP to Q_B	"	25	"
	t_{PHL5}	"	170																	CP to Q_B	"	36	"
	t_{PLH5}	"	171																	CP to Q_C	"	26	"
	t_{PHL5}	"	172																	CP to Q_C	"	36	"
	t_{PLH5}	"	173																	CP to Q_D	"	26	"
	t_{PHL5}	"	174																	CP to Q_D	"	36	"
	t_{PHL7}	"	175																	ET to RC	"	32	"
	t_{PLH7}	"	176																	ET to RC	"	28	"
	t_{PHL11}	"	177																	U/$\overline{D}$ to RC	"	37	"
	t_{PLH9}	"	178																	U/$\overline{D}$ to RC	"	32	"
	t_{PHL7}	"	179																	ET to RC	"	32	"
	t_{PLH7}	"	180																	ET to RC	"	28	"
	t_{PHL11}	"	181																	U/$\overline{D}$ to RC	"	37	"
	t_{PLH9}	"	182																	U/$\overline{D}$ to RC	"	32	"
	F_{MAX}	"	183																	CP to Q_n	25		MHz
	F_{MAX}	"	184																	CP to Q_n	25		MHz
11	Same tests, conditions and limits as for subgroup 10 except T_C = -55°C and V_{CC} = 4.5 for F_{MAX}.																						

Same conditions as for subgroup 9.

See footnotes at end of device types 05 and 06.

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